

-- b. Create a view, "Lab001.OrderDetail" based on the user-friendly

-- order details query below

```
CREATE OR ALTER VIEW Lab001.OrderDetail AS
```

```
SELECT oh.OrderDate,
```

```
oh.OrderID,
```

```
cu.FirstName,
```

```
cu.LastName,
```

```
cu.AddressLine1,
```

```
cu.City,
```

```
cu.RegionAbbr,
```

```
cu.PostalCode,
```

```
p.ProductName,
```

```
ol.Quantity,
```

```
p.ProductPrice
```

```
FROM Lab001.customer cu, Lab001.OrderHeader oh, Lab001.OrderLine ol,
```

```
Lab001.product p
```

```
WHERE cu.CustomerID=oh.CustomerID AND
```

```
oh.OrderID=ol.OrderID AND
```

```
ol.ProductID=p.ProductID
```

```
GO
```

-- c. Select all rows and columns from Lab001.OrderDetail and take a screenshot

```
SELECT * FROM Lab001.OrderDetail;
```

Results (1)	Messages										Open in New Tab	
	OrderDate	OrderID	FirstName	LastName	AddressLine1	City	RegionAbbr	PostalCode	ProductName			
1	2022-10-22	1003	Brian	Borg	100 College	SF	CA	17118	Printer			
2	2022-10-22	1004	Jane	Smith	120 University	Salt Lake City	UT	84102	Stapler			
3	2022-10-22	1004	Jane	Smith	120 University	Salt Lake City	UT	84102	Shredder			
4	2022-10-24	1005	Josh	Turburn	250 North	Salt Lake City	UT	84108	Desk			
5	2022-10-24	1006	Bill	Doll	300 West	Salt Lake City	UT	84107	Desk			
6	2022-10-24	1006	Bill	Doll	300 West	Salt Lake City	UT	84107	Office Chair			
7	2022-10-24	1006	Bill	Doll	300 West	Salt Lake City	UT	84107	Filing Cabinet			
8	2022-10-27	1007	Joyce	French	200 Main	Tucson	AZ	45102	Laptop Comput			
9	2022-10-30	1008	Jennifer	English	250 State	Tucson	AZ	45112	Printer			
10	2022-10-30	1008	Jennifer	English	250 State	Tucson	AZ	45112	Shredder			
11	2022-11-01	1009	Mary	Lee	200 South	Salt Lake City	UT	84102	Desk			
12	2022-11-01	1009	Mary	Lee	200 South	Salt Lake City	UT	84102	Filing Cabinet			

-- c. Increase price for product by \$10

UPDATE Lab001.product

SET ProductPrice = ProductPrice + 10.00

WHERE ProductID = 101

GO

-- d. Verify trigger has worked

SELECT *

FROM Lab001.ProductPriceAudit

GO

Results (1)		Messages	
	UpdateDate	OldValue	NewValue
1	2026-03-07	9528.7242	9538.7242

ALTER TABLE Lab001.ProductPriceAudit

ADD price_change_percentage NUMERIC(10,3);

GO

-- f. Update the Lab001.AuditPriceChangeTrigger trigger to add the price percentage change

CREATE OR ALTER TRIGGER Lab001.AuditPriceChangeTrigger

ON [Lab001].[product]

FOR UPDATE

AS

BEGIN

SET NOCOUNT ON;

DECLARE @oldValue MONEY;

DECLARE @newValue MONEY;

DECLARE @percentageChange NUMERIC(10,3);

SELECT @oldValue = ProductPrice FROM deleted;

SELECT @newValue = ProductPrice FROM inserted;

```

SET @percentageChange =
    CASE
        WHEN @oldValue = 0 THEN NULL
        ELSE ((@newValue - @oldValue) * 100) / @oldValue
    END;
INSERT INTO Lab001.ProductPriceAudit
(
    UpdateDate,
    OldValue,
    NewValue,
    price_change_percentage
)
VALUES
(
    GETDATE(),
    @oldValue,
    @newValue,
    @percentageChange
);
END
GO

```

-- g. Increase price for product by 10%

```

UPDATE Lab001.product
SET ProductPrice = ProductPrice * 1.1 -- 10% price increase
WHERE ProductID = 101
GO

```

-- h. Verify the trigger has worked by selecting all rows and

-- columns from Lab001.ProductPriceAudit and take a screenshot

SELECT *

FROM Lab001.ProductPriceAudit

GO

Results (1)		Messages						
	UpdateDate	↕	OldValue	↕	NewValue	↕	price_change_percentage	↕
1	2026-03-07		9528.7242		9538.7242		NULL	
2	2026-03-07		9538.7242		10492.5966		10.000	

-- c. Create a new function called Lab001.calculate_discounted_percent that takes

-- a product id and a new price and returns the discount percentage

CREATE OR ALTER FUNCTION Lab001.calculate_discounted_percent(

@productID INT,

@newPrice NUMERIC(10,3))

RETURNS MONEY

AS

BEGIN

DECLARE @returnValue AS MONEY;

SELECT @returnValue = ((ProductPrice - @newPrice)/ProductPrice)

FROM [Lab001].[product]

WHERE [Lab001].[product].ProductID = @productID

RETURN(@returnValue)

END

GO

-- For example, if ProductID 103 was discounted to \$29.50 it would return approx 0.81

-- d. Run the function discounting ProductID 103 to \$29.50 and take a screenshot
SELECT Lab001.calculate_discounted_percent(103, 29.50) AS DiscountedPercent
GO

Results (2)		Messages
	DiscountedPrice	↕ ⌵
1	994.36	
	DiscountedPercent	↕ ⌵
1	0.8145	